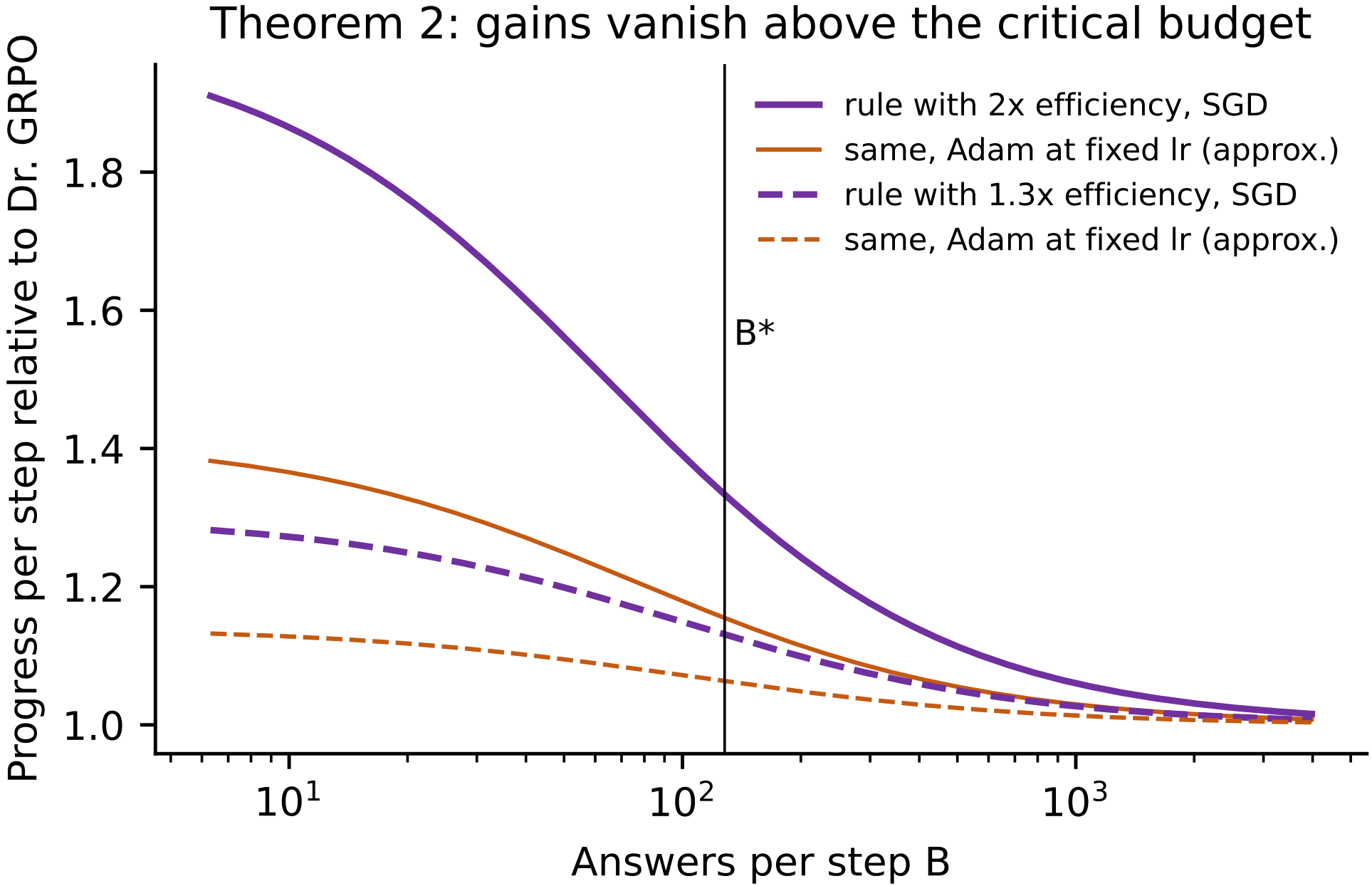
Theorem 2: gains vanish above the critical budget
Progress per step relative to Dr. GRPO
1.8
1.6
1.4
1.2
1.0
B*
rule with 2x efficiency, SGD
same, Adam at fixed lr (approx.)
rule with 1.3x efficiency, SGD
same, Adam at fixed lr (approx.)
10^1
10^2
10^3
Answers per step B